BABU BANARASI DAS UNIVERSITY

2026 - 2027



SCHOOL OF COMPUTER APPLICATIONS

No SQL Project On

Student Management System Using MongoDB

Submitted to:

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Date: 25th September

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1. CREATE DATABASE

MongoDB Query:

use Students

Explanation: Creates or switches to the Students database.

MongoDB Shell — Students

```
> MONGOSH
use Students
switched to db Students
```

2. CREATE COLLECTION

MongoDB Query:

db.createCollection("students")

Explanation: Creates a collection named students.

MongoDB Shell — Students

```
> MONGOSH
db.createCollection("students")
{ ok: 1 }
```

3. INSERT 50 STUDENTS

MongoDB Query:

db.students.insertMany([...])

Explanation: Inserts 50 student documents into the collection.

MongoDB Shell — Students

```
> MONGOSH
db.students.insertMany([
  { _id: ObjectId('6f2c4b81d93a07e5c61fa300'), roll: 1, name: 'Pranav', age: 19, marks: 57, city: 'Mathura' }
  { _id: ObjectId('6f2c4b81d93a07e5c61fa301'), roll: 2, name: 'Neha', age: 21, marks: 51, city: 'Kanpur' }
  { _id: ObjectId('6f2c4b81d93a07e5c61fa302'), roll: 3, name: 'Aditya', age: 20, marks: 93, city: 'Ghaziabad' }
  // ... 47 more student documents
])
{ acknowledged: true, insertedCount: 50 }
```

4. DISPLAY ALL STUDENTS

MongoDB Query:

db.students.find()

Explanation: Retrieves all documents from the students collection.

MongoDB Shell — Students

```
> MONGOSH
db.students.find()
{ _id: ObjectId('6f2c4b81d93a07e5c61fa300'), roll: 1, name: 'Pranav', age: 19, marks: 57, city: 'Mathura' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa301'), roll: 2, name: 'Neha', age: 21, marks: 51, city: 'Kanpur' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa302'), roll: 3, name: 'Aditya', age: 20, marks: 93, city: 'Ghaziabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa303'), roll: 4, name: 'Kavya', age: 20, marks: 51, city: 'Moradabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa304'), roll: 5, name: 'Harsh', age: 18, marks: 62, city: 'Agra' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa305'), roll: 6, name: 'Muskan', age: 19, marks: 90, city: 'Agra' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa306'), roll: 7, name: 'Vivek', age: 18, marks: 53, city: 'Moradabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa307'), roll: 8, name: 'Shruti', age: 19, marks: 88, city: 'Jhansi' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa308'), roll: 9, name: 'Naman', age: 18, marks: 62, city: 'Jhansi' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa309'), roll: 10, name: 'Ananya', age: 18, marks: 71, city: 'Jhansi' }
Type "it" for more (40 more documents)
```

5. EQUAL TO --- $eq

MongoDB Query:

db.students.find({ marks: { $eq: 88 } })

Explanation: Finds students whose marks are exactly 88.

MongoDB Shell — Students

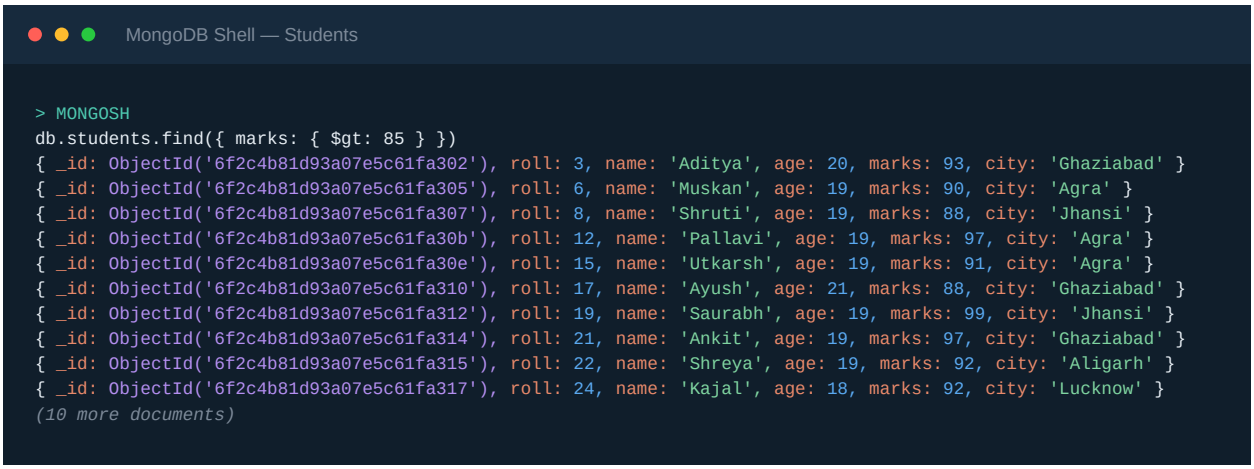
```
> MONGOSH
db.students.find({ marks: { $eq: 88 } })
{ _id: ObjectId('6f2c4b81d93a07e5c61fa307'), roll: 8, name: 'Shruti', age: 19, marks: 88, city: 'Jhansi' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa310'), roll: 17, name: 'Ayush', age: 21, marks: 88, city: 'Ghaziabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa31c'), roll: 29, name: 'Piyush', age: 20, marks: 88, city: 'Agra' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa328'), roll: 41, name: 'Ritesh', age: 19, marks: 88, city: 'Agra' }
```

6. GREATER THAN --- $gt

MongoDB Query:

db.students.find({ marks: { $gt: 85 } })

Explanation: Finds students whose marks are greater than 85.

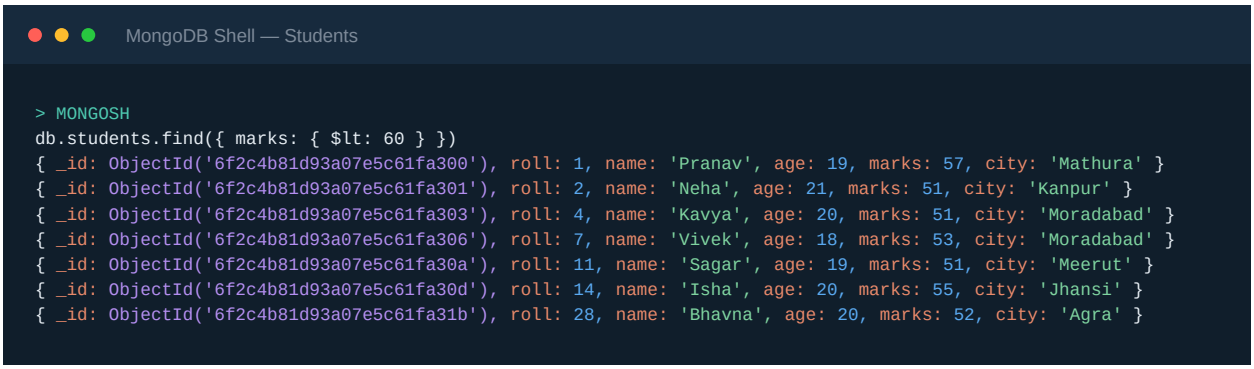


7. LESS THAN --- $lt

MongoDB Query:

db.students.find({ marks: { $lt: 60 } })

Explanation: Finds students whose marks are less than 60.



8. GREATER THAN OR EQUAL TO --- $gte

MongoDB Query:

db.students.find({ marks: { $gte: 90 } })

Explanation: Finds students whose marks are 90 or above.

MongoDB Shell — Students

```
> MONGOSH
db.students.find({ marks: { $gte: 90 } })
{ _id: ObjectId('6f2c4b81d93a07e5c61fa302'), roll: 3, name: 'Aditya', age: 20, marks: 93, city: 'Ghaziabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa305'), roll: 6, name: 'Muskan', age: 19, marks: 90, city: 'Agra' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa30b'), roll: 12, name: 'Pallavi', age: 19, marks: 97, city: 'Agra' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa30e'), roll: 15, name: 'Utkarsh', age: 19, marks: 91, city: 'Agra' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa312'), roll: 19, name: 'Saurabh', age: 19, marks: 99, city: 'Jhansi' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa314'), roll: 21, name: 'Ankit', age: 19, marks: 97, city: 'Ghaziabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa315'), roll: 22, name: 'Shreya', age: 19, marks: 92, city: 'Aligarh' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa317'), roll: 24, name: 'Kajal', age: 18, marks: 92, city: 'Lucknow' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa319'), roll: 26, name: 'Priya', age: 19, marks: 94, city: 'Ghaziabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa320'), roll: 33, name: 'Abhay', age: 20, marks: 90, city: 'Ghaziabad' }
(5 more documents)
```

9. LESS THAN OR EQUAL TO --- $lte

MongoDB Query:

db.students.find({ marks: { $lte: 70 } })

Explanation: Finds students whose marks are 70 or below.

MongoDB Shell — Students

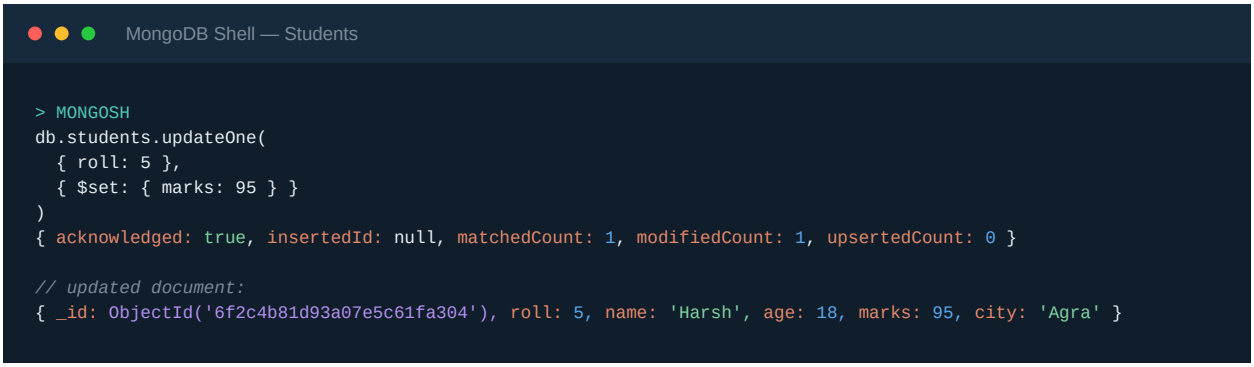
```
> MONGOSH
db.students.find({ marks: { $lte: 70 } })
{ _id: ObjectId('6f2c4b81d93a07e5c61fa300'), roll: 1, name: 'Pranav', age: 19, marks: 57, city: 'Mathura' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa301'), roll: 2, name: 'Neha', age: 21, marks: 51, city: 'Kanpur' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa303'), roll: 4, name: 'Kavya', age: 20, marks: 51, city: 'Moradabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa304'), roll: 5, name: 'Harsh', age: 18, marks: 62, city: 'Agra' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa306'), roll: 7, name: 'Vivek', age: 18, marks: 53, city: 'Moradabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa308'), roll: 9, name: 'Naman', age: 18, marks: 62, city: 'Jhansi' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa30a'), roll: 11, name: 'Sagar', age: 19, marks: 51, city: 'Meerut' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa30d'), roll: 14, name: 'Isha', age: 20, marks: 55, city: 'Jhansi' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa316'), roll: 23, name: 'Tushar', age: 19, marks: 67, city: 'Meerut' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa31b'), roll: 28, name: 'Bhavna', age: 20, marks: 52, city: 'Agra' }
(6 more documents)
```

10. UPDATE OPERATION

MongoDB Query:

```
db.students.updateOne(
  { roll: 5 },
  { $set: { marks: 95 } }
)
```

Explanation: Updates the marks to 95 of the student with roll number 5.



11. DELETE ONE OPERATION

MongoDB Query:

```
db.students.deleteOne({ roll: 10 })
```

Explanation: Deletes one document matching roll number 10.

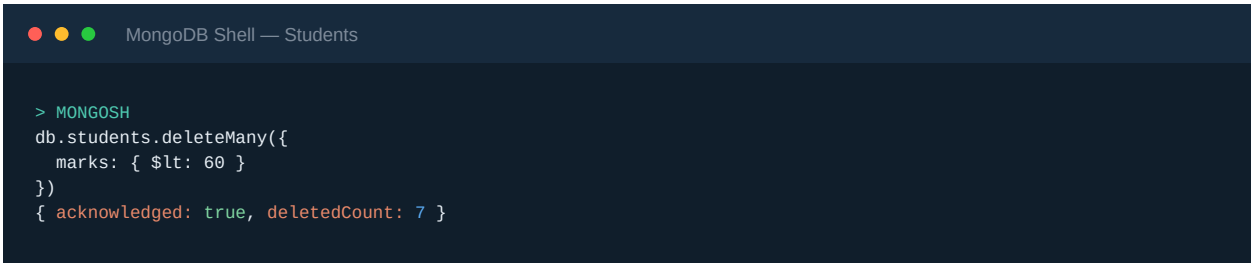


12. DELETE MANY OPERATION

MongoDB Query:

```
db.students.deleteMany({
  marks: { $lt: 60 }
})
```

Explanation: Deletes all students having marks below 60.



13. AND OPERATION --- $and

MongoDB Query:

```
db.students.find({
  $and: [
    { marks: { $gt: 80 } },
    { age: 19 }
  ]
})
```

Explanation: Finds students satisfying both conditions.

MongoDB Shell — Students

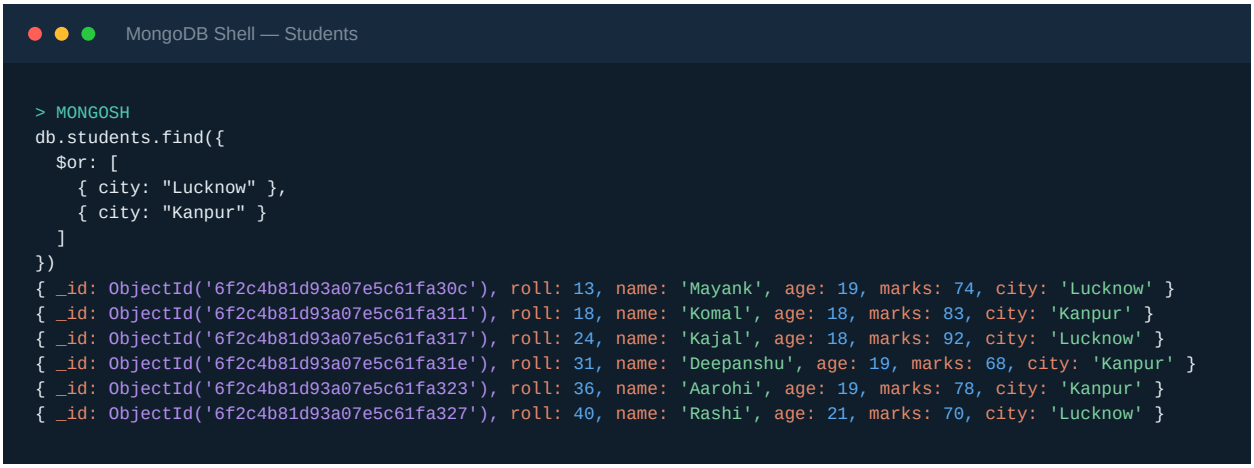
```
> MONGOSH
db.students.find({
  $and: [
    { marks: { $gt: 80 } },
    { age: 19 }
  ]
})
{ _id: ObjectId('6f2c4b81d93a07e5c61fa305'), roll: 6, name: 'Muskan', age: 19, marks: 90, city: 'Agra' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa307'), roll: 8, name: 'Shruti', age: 19, marks: 88, city: 'Jhansi' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa30b'), roll: 12, name: 'Pallavi', age: 19, marks: 97, city: 'Agra' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa30e'), roll: 15, name: 'Utkarsh', age: 19, marks: 91, city: 'Agra' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa312'), roll: 19, name: 'Saurabh', age: 19, marks: 99, city: 'Jhansi' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa314'), roll: 21, name: 'Ankit', age: 19, marks: 97, city: 'Ghaziabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa315'), roll: 22, name: 'Shreya', age: 19, marks: 92, city: 'Aligarh' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa319'), roll: 26, name: 'Priya', age: 19, marks: 94, city: 'Ghaziabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa324'), roll: 37, name: 'Kartik', age: 19, marks: 96, city: 'Noida' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa326'), roll: 39, name: 'Vansh', age: 19, marks: 93, city: 'Mathura' }
(4 more documents)
```

14. OR OPERATION --- $or

MongoDB Query:

```
db.students.find({
  $or: [
    { city: "Lucknow" },
    { city: "Kanpur" }
  ]
})
```

Explanation: Finds students from Lucknow or Kanpur.



15. SORT OPERATION (Ascending)

MongoDB Query:

```
db.students.find().sort({ marks: 1 })
```

Explanation: Sorts students by marks from lowest to highest. Use 1 for ascending order.



15. SORT OPERATION (Descending)

MongoDB Query:

db.students.find().sort({ marks: -1 })

Explanation: Sorts students by marks from highest to lowest. Use -1 for descending order.

MongoDB Shell — Students

```
> MONGOSH
db.students.find().sort({ marks: -1 })
{ _id: ObjectId('6f2c4b81d93a07e5c61fa312'), roll: 19, name: 'Saurabh', age: 19, marks: 99, city: 'Jhansi' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa30b'), roll: 12, name: 'Pallavi', age: 19, marks: 97, city: 'Agra' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa314'), roll: 21, name: 'Ankit', age: 19, marks: 97, city: 'Ghaziabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa324'), roll: 37, name: 'Kartik', age: 19, marks: 96, city: 'Noida' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa304'), roll: 5, name: 'Harsh', age: 18, marks: 95, city: 'Agra' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa319'), roll: 26, name: 'Priya', age: 19, marks: 94, city: 'Ghaziabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa302'), roll: 3, name: 'Aditya', age: 20, marks: 93, city: 'Ghaziabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa326'), roll: 39, name: 'Vansh', age: 19, marks: 93, city: 'Mathura' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa32b'), roll: 44, name: 'Simran', age: 18, marks: 93, city: 'Meerut' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa330'), roll: 49, name: 'Dhruv', age: 19, marks: 93, city: 'Mathura' }
(32 more documents)
```

16. LIMIT OPERATION

MongoDB Query:

db.students.find().limit(5)

Explanation: Displays only the first five documents.

MongoDB Shell — Students

```
> MONGOSH
db.students.find().limit(5)
{ _id: ObjectId('6f2c4b81d93a07e5c61fa302'), roll: 3, name: 'Aditya', age: 20, marks: 93, city: 'Ghaziabad' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa304'), roll: 5, name: 'Harsh', age: 18, marks: 95, city: 'Agra' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa305'), roll: 6, name: 'Muskan', age: 19, marks: 90, city: 'Agra' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa307'), roll: 8, name: 'Shruti', age: 19, marks: 88, city: 'Jhansi' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa308'), roll: 9, name: 'Naman', age: 18, marks: 62, city: 'Jhansi' }
```

17. SEARCH BY CITY

MongoDB Query:

db.students.find({ city: "Lucknow" })

Explanation: Finds all students whose city is Lucknow.

MongoDB Shell — Students

```
> MONGOSH
db.students.find({ city: "Lucknow" })
{ _id: ObjectId('6f2c4b81d93a07e5c61fa30c'), roll: 13, name: 'Mayank', age: 19, marks: 74, city: 'Lucknow' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa317'), roll: 24, name: 'Kajal', age: 18, marks: 92, city: 'Lucknow' }
{ _id: ObjectId('6f2c4b81d93a07e5c61fa327'), roll: 40, name: 'Rashi', age: 21, marks: 70, city: 'Lucknow' }
```

18. COUNT DOCUMENTS

MongoDB Query:

db.students.countDocuments()

Explanation: Counts documents currently present in the collection (performed after the delete operations).

